## Machine learning and mathematics

Many machine learning algorithms basically attempt to approximate an unknown mathematical function, and most of these algorithms are mathematical in nature. As data contains randomness and uncertainty, we'll have to apply concepts from probability and statistics. In fact, machine learning algorithms are so dependent on concepts from statistics that many people refer to machine learning as statistical learning. Apart from statistics, another important branch of mathematics that's very much in use is linear algebra. Concepts of matrices, solutions for systems of equations and optimisation algorithms — all play important roles in machine learning.

## Machine learning and Big Data

These days, machine learning is closely associated with Big Data. Big Data refers to large volumes of stored data, and it always makes sense to use. Let's look at a specific case to understand what exactly I mean. Let's suppose that you've been collecting customers' transaction data on an e-commerce platform. The volume of data collected can easily grow and eventually reach a level at which it gets categorised as Big Data, along with other data you're storing. Now, there are primary uses of this data — to check the status of that customer order, to calculate profits/losses of the business, for accounting purposes and some other routine operational/technical purposes. Apart from these, businesses also want to make use of this data in a unique way. Some people have a specific word for this – analytics. While straightforward arithmetic can give answers on profits/losses, total expenditure, total inventory, etc, and slightly advanced data manipulation can *slice and dice* the information available, we would still want to go beyond this basic reporting and identify patterns that are not obvious. We may also want to build predictive models.

## Different machine learning algorithms

There are basically two types of machine learning –- supervised learning and unsupervised learning. Supervised learning refers to data with labels, examples of which are shown below:

| Day | Temperature at 12 pm | Wind speed | Wind direction | Rain in the evening |
|---|---|---|---|---|
| 1 | 30°C | 3 m/s | East | Yes |
| 2 | 35°C | 10 m/s | West | No |
| 3 | 37°C | 7 m/s | North-east | No |
| 4 | 34°C | 8 m/s | North | No |

Let's assume that we have to predict if it rains in the evening, using temperature and wind data. Whether it rains or not is stored in the column *Rain in the evening*, which becomes the label. Here, temperature, wind speed and wind direction become *predictors* or *input variables* and *Rain in the evening* becomes the output variable.

The algorithms that learn from such data are called supervised learning algorithms. While some data can be extracted/generated automatically, like in a machine log, often it may have to be labelled manually, which would increase the cost of data acquisition.

Examples for supervised learning algorithms include linear regression algorithms, Naïve-Bayesan algorithms, k-nearest neighbour, etc.

If the data doesn't have labels, it becomes an unsupervised learning algorithm. Some examples are k-means clustering, principal component analysis, etc.

We can also classify machine learning algorithms using a different logic — regression algorithms and classification algorithms. Regression algorithms are machine learning algorithms that actually predict a 'number' — like the following day's temperature, the stock market's closing index, etc. Classification algorithms are those that can classify an input, like whether it will rain or not; if the stock market will close positive or negative; if it is disease x, disease y, or disease z, and so on.

## Tools of machine learning

It's important to understand and appreciate the fact that machine learning algorithms are basically mathematical algorithms and we can implement them in any language we like. It can be C, PHP, Java, Python, or even JavaScript. But the one I personally like and use a lot is the R statistical language. There are many popular machine learning modules or libraries in different languages. Weka is powerful machine learning or data mining software written in Java and is very popular. Scikit-learn is popular among Python developers. One can also go for the Orange machine learning toolbox available in Python. While R comes with many basic statistical packages or modules, there are different packages like e1071, Randomforest, Ann, etc, depending on different machine learning algorithms. Apache Mahout is a scalable machine learning algorithm typically for Big Data systems. While Weka is really powerful, it has some licence issues for commercial use. You can also use Java-ML, a Java machine learning algorithm. Though development appears to have stopped, it is a decent library worth trying and with enough documentation/tutorials to get started easily. I would recommend people with advanced needs to explore the deep learning algorithms. END

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